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Combined effects of deterministic and statistical structure on high-frequency regional seismograms

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SUMMARY

Radiative transport modelling can combine the effects of both large-scale (deterministic) and the small-scale (statistical) structure on the coda envelopes of high-frequency regional seismograms. We describe a computer code to implement radiative transport modelling that propagates packets of seismic body wave energy along ray paths through large-scale deterministic 3-D structure, including the effects of velocity gradients, intrinsic attenuation, source radiation pattern and multiple scattering by layer boundaries and small-scale heterogeneities specified by a heterogeneity spectrum. The spatial distribution of these energy packets can be displayed as time snapshots to aid in the understanding of regional phase propagation or displayed as a coda envelope by summing at receiver bins. These techniques are applied to earthquakes and explosions recorded in the Lop Nor, China region to model observed narrow band passed seismic codas in the 1–4 Hz band. We predict that source discriminants in this region based on P/Lg amplitude ratios will best separate earthquake and explosion populations at frequencies 2 Hz and higher.

Key words: Coda waves; Earthquake monitoring and test-ban treaty verification; Seismic attenuation; Wave propagation; Wave scattering and diffraction.

1 INTRODUCTION

1.1 The importance and limitations of very high frequency seismogram modelling

Efficient modelling of high-frequency regional seismograms is important to both the assessment of earthquake hazards and to nuclear treaty verification. Earthquake damage correlates most strongly with ground motion in the 0.5–10 Hz frequency band (Trifunac & Brady 1975). The spectrum of an earthquake seismogram in the 2 to 6 Hz band can discriminate it from that of an explosion or quarry blast better than either its estimated source mechanism or source depth (Walter *et al.* 1995, 2007; Allmann *et al.* 2008).

Given the importance of high frequencies to hazards estimation and seismic monitoring, it would be desirable to model high frequency regional seismograms with enough computational efficiency that many structural and source models could be routinely tested and compared against observations within several hours or less. Unfortunately, even with advances in parallel computation and increased access to large clusters of processors, numerical modelling in 3-D Earth models rarely exceeds ranges of several hundred wavelengths for computing times less than one day. At 5 Hz this range is typically less than 100–200 km, which is valuable to predictions of earthquake strong ground motion, but limited in value to nuclear verification where monitoring stations may be 1000 km or further from regions of interest in Eurasia. Even if the computational inefficiency can be overcome, as frequency increases it becomes increasingly difficult to obtain perfect knowledge of small-scale structure needed to predict a seismogram of an earthquake or an explosion wiggle for wiggle.

1.2 Limitations of empirical Green's functions and path calibrations

Several approaches are available for handling imperfect knowledge of Earth structure for high frequency coda modelling. Strong ground motion modelling often employs empirical Green's functions obtained from recordings of previous earthquakes in a region at similar ranges of interest to simulate the effects of scattering by unknown small-scale heterogeneity (Hartzell 1978). Sophisticated interpolating (e.g. kriging, Fan *et al.* 2002) has been used predict the behaviour of the amplitudes of regional seismic phases over unsampled new paths using the amplitudes of these phases recorded over other paths within a region. These techniques may address the effects of unknown small-scale (<10 km heterogeneity) if it is relatively uniform over a large region. Resolvable sharp gradients in Moho depth and deep sedimentary basins, however, can still strongly affect azimuthal variations in the character and detection of regional seismic phases (e.g. Cao & Muirhead 1993; Sens-Schönfelder *et al.* 2009). These resolvable, deterministic, structures make simpler 1-D

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structural models, empirical Green's functions, and path interpolation unreliable tools for predicting waveform behaviour.

1.3 An efficient method of incorporating multiple scales of heterogeneity

To address this problem, this paper extends a radiative transport algorithm that combines ray theory to treat the effects of known large-scale structure together with the effects of multiple-scattering by perturbations of the large-scale structure. The following sections and an appendix describe the implementation of the algorithm and its application to high-frequency regional seismograms in the vicinity of the Lop Nor, China test site.

2 TECHNICAL APPROACH

2.1 Radiative transport

The term *coda* for the envelope of complex arrivals following direct *P* and *S* waves at local and regional distances originated with work by Aki (1969), who recognized their origin to be due to multiple scattering by small-scale (relative to wavelength) heterogeneities in the crust and lithosphere (Aki & Chouet 1975). The radiative transport algorithm models these coda envelopes by simulating the diffusive transport of elastic energy through processes of multiple scattering in a medium having random fluctuations in elastic properties. Its earliest and most extensive applications are to the propagation of electromagnetic waves, especially to the description of stellar images and scintillation (Chandrasekhar 1960). Gusev & Abubakirov (1987) extended applications of radiative transport to elastic scattering using a Monte Carlo technique for tracking multiply scattered particles of elastic energy. Extensions to realistic Earth models followed in the work of Hoshiba (1997), who included particle interactions with layer boundaries, Yoshimoto (2000), who included effects of velocity gradients within layers, and Margerin *et al.* (2000), who elaborated treatment of *S* wave polarization. An early significant result of work with radiative transport is that multiple scattering often becomes the dominant mechanism controlling coda shapes at frequencies above 5 Hz on Earth, explaining a spindle-shaped, lunar-like, seismic coda observed in this frequency band (Wu 1985). Hoshiba (1993) developed an analysis method now in common use for measuring the relative contributions of intrinsic versus scattering attenuation from the shape of seismic codas. Comprehensive reviews of seismic radiative transport modelling include the text by Sato *et al.* (2012) and an AGU monograph chapter by Margerin (2005). Przybilla *et al.* (2009) have compared the method against finite-difference synthetics in 3-D, validating body wave coda predictions of the radiative transport method.

Following Shearer & Earle (2008), an algorithm to implement radiative transport for computing Monte Carlo trajectories of energy particles can be summarized in a few of simple steps:

(i) Choose a heterogeneity power spectrum for the medium (Fig. 1). The heterogeneity power spectrum is the Fourier transform of the spatial autocorrelation of small-scale heterogeneity of seismic velocities and density.

(ii) From the heterogeneity spectrum and background velocities, calculate the scattering coefficient, $g(\theta, \phi)$, which is the scattering power per unit volume at a deflection angle given by θ and ϕ. Four separate g functions are computed to capture the possibility of scattering induced mode conversions between *P* and *S* waves.

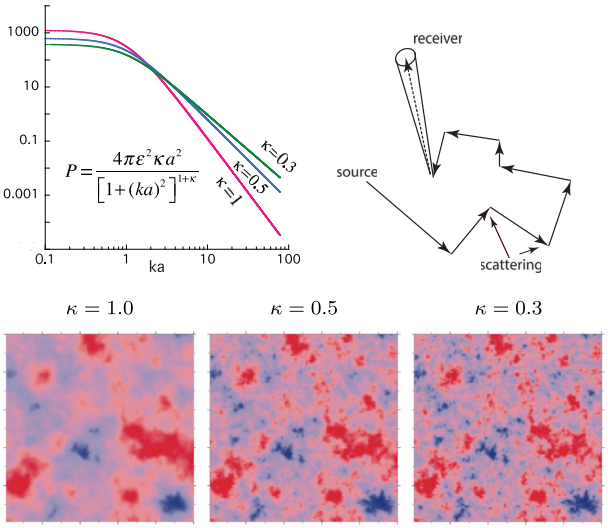


Figure 1. Top right: random walk in which deterministic ray paths conveying packets of elastic energy are interrupted by scattering events in a heterogeneous medium. Top left: 2-D von Kármán heterogeneity spectrum as a function of correlation wavenumber k; spectral power is flat up to $k_{\text{corner}} \sim 1/a$, after which smaller spatial scales decay with fall-off rate controlled by Hurst parameter κ. Bottom: example spatial realizations of heterogeneity spectra for various κ at a magnification level of ten scale lengths a.

The g functions are calculated separately for *P*-to-*P*, *P*-to-*S*, *S*-to-*S*, and *S*-to-*P* scattering (Sato *et al.* 2012).

(iii) Calculate mean free paths by averaging each of the four g function over all angles to form total scattering coefficients g_0, and then sum on initial wave mode as follows: $g_0^{(P)} = g_0^{P\to P} + g_0^{P\to S} = 1/\ell_{\text{P}}$, where length ℓ_{P} is the mean free path of a *P* wave, and similarly for *S* waves.

(iv) Compute a random path length r to represent one trajectory segment of the random walk by assuming r is an exponentially distributed random number having a mean value ℓ_{P} for *P* waves and ℓ_{S} for *S* waves. For a *P* wave, compute $r_{\text{P}} = -\ell_{\text{P}} \ln(x)$, where x is a random number between 0 and 1, and advance the energy particle this distance along a trajectory computed by ray theory. (Note: If mean free paths are spatially varying due to smooth or discrete variation of background velocities or heterogeneity spectrum, and if r exceeds a tolerance over which the mean free path may acceptably vary, then it may be necessary to subdivide the propagation step into smaller path lengths and recompute r at each step.)

(v) Adjust ray tangent and polarization mode by interpreting the Born scattering coefficients $g^{X\to P}(\theta, \phi)$ and $g^{X\to S}(\theta, \phi)$ (for X representing initial wave mode) as probability densities determining the relative likelihood of scattering into each deflection angle and wave mode, and continue computing the path of the scattered energy packet by iterating step 4 above. The evolution of *S* polarization along a path is handled by assigning *S* polarity as an angle in a local *SV* versus *SH* coordinate system (Shearer & Earle 2004). This is an approximation that ignores the possibility of elliptical *S* polarization induced by *SV* interaction with interfaces, which can be remedied using the approaches of Margerin *et al.* (2000) and Bal & Moscoso (2000).

2.2 Advantages of radiative transport

The principle advantage of radiative transport is the ability to simulate the effects of small-scale structure without the need for model

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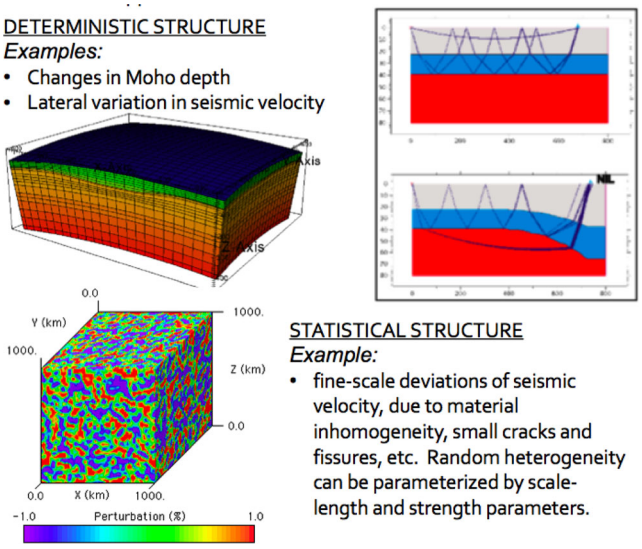


Figure 2. Top: examples of known (deterministic) large-scale heterogeneity and its effects on *S* and *Lg* waves (multiple critically reflected *S* waves at the Moho). Bottom: example of a realization of 3-D small-scale (statistically described) heterogeneity.

meshes that are dense enough to explicitly describe that structure. Instead, the model mesh describes only the large-scale, long-wavelength equivalent ($ka \gg 1$) background medium, against which small-scale heterogeneities are assumed as a random perturbation field. The small-scale heterogeneities are described in the aggregate via a small number of statistical parameters, rather than describing the perturbation field explicitly in fine detail.

2.3 Construction of models

Earth models consist of a background deterministic component, consisting of first-order discontinuities with topography separating layers having 3-D variation of velocities and densities interpolated over a course grid, and a perturbed statistical component, specified by a heterogeneity spectrum parametrized by the power of velocity and density fluctuation and a small number of spectral shape parameters informing the functional dependence of the power spectrum on wavenumber (Figs 1 and 2).

Since Frankel and Clayton's (1986) work demonstrating the effects of small-scale heterogeneity on the seismic wavefield, significant advances have occurred in statistical characterization and understanding of such heterogeneity. Goff *et al.* (1994) described a procedure by which models having the spatial statistics of polycrystalline or multi-modal assemblages of rocks can be generated. Work beginning with Levander *et al.* (1994) and Pullammanappallil *et al.* (1997) makes it possible to formulate statistical models for common sedimentary and metamorphic formations. Model statistics can be modified as needed to reproduce the three-component behaviour of the coda of regional seismic phases. Strong emphasis is placed on correctly characterizing the statistics of small-scale heterogeneities in the upper, highly heterogeneous, 10 km of the earth, which may correlate with the depth to a brittle to ductile transition in the crust (Rachman & Chung 2016). Small-scale statistics affect the partitioning of *P* and *S* energy on the three-components of motion by the effects of scattering near the source and receiver. These small-scale statistics can strongly affect the performance of discriminants tuned to differences in sources occurring at the depths of contained nuclear tests.

Lateral variations in crustal thickness, basin depths, mountain roots, and lateral tectonic transitions also significantly affect the phases used for discrimination and detection. One example is the study by Pedersen *et al.* (1998), who explain anomalous Rayleigh to Love mode conversion from Lop Nor explosions by changes in crustal thickness at the boundary of the Tarim Basin and Tian Shan mountain belt. Moho topography and basin thickness can also strongly affect the propagation of *Lg* (Cormier & Anderson 2004). The scale of these types of lateral structural variations is often large enough to be resolvable by local and regional reflection and refraction experiments, gravity and magnetic data, regionalization by surficial geology, and global surface wave inversions. Hence, we refer to these types of structures as deterministic. The types of data used to infer deterministic structure are collected at widely different spatial scales, presenting a challenge to the parametrization of a 3-D model appropriate for a region surrounding a particular seismic station. The parametrization should be flexible enough to be specified at high resolution where data is available and at lower resolution where it is not. Resolution should be high to describe features important to regional wave propagation, such as Moho and basin topography, but can be lower near interfaces having smaller velocity contrasts and lower with increasing depth in the mantle, where heterogeneity power decreases.

3 SOFTWARE PACKAGE

3.1 Implementation and relation to existing codes

Our code RADIATIVE3D is a tool for seismic radiative transport modelling with flexible Earth model and moment-tensor source input. The simulation engine includes deterministic ray tracing for longitudinal and shear propagation modes in linear velocity gradients, stochastic scattering controlled by heterogeneity parameters, reflection and transmission handling at sharp boundaries, and attenuation modelling from internal friction. Visualization and post-processing tools have been created to produce seismic envelopes, traveltime curves, and 3-D body-wave visualizations of energy propagation from RADIATIVE3D output. Additional capabilities for source and model input are described in Appendix A1. RADIATIVE3D is made available to the scientific community as a free and open-source software product (http://rainbow.phys.uconn.edu/geowiki/Radiative3D). RADIATIVE3D is a new code that takes major inspiration from two prior codes, namely Raytrace3D (Menke 2005) and PSPhonon Shearer & Earle (2004). Raytrace3D solves ray propagation in 3-D velocity models, and PSPhonon performs radiative transport, including scattering, in 1-D layered-Earth models. Our code performs both tasks in a full 3-D model.

In our implementation, elastic energy is considered to originate from a source event at a fixed location and time. Energy is then propagated as a discrete packet or bundle, which we refer to as a phonon (in analogy to the particle representation of light as a stream of photons). A phonon's path through an Earth model is determined by a combination of ray theory to handle the deterministic propagation through the composite medium background structure (large-scale structure, capturing broad variations in elastic properties), and by scattering theory, which is the stochastic handling of scattering due to small-scale heterogeneities, assumed as perturbations to the background structure. A scattering event is, in essence, an interruption and randomization of a particle's otherwise deterministic progress. With a sufficient number of phonons emitted from the source event,

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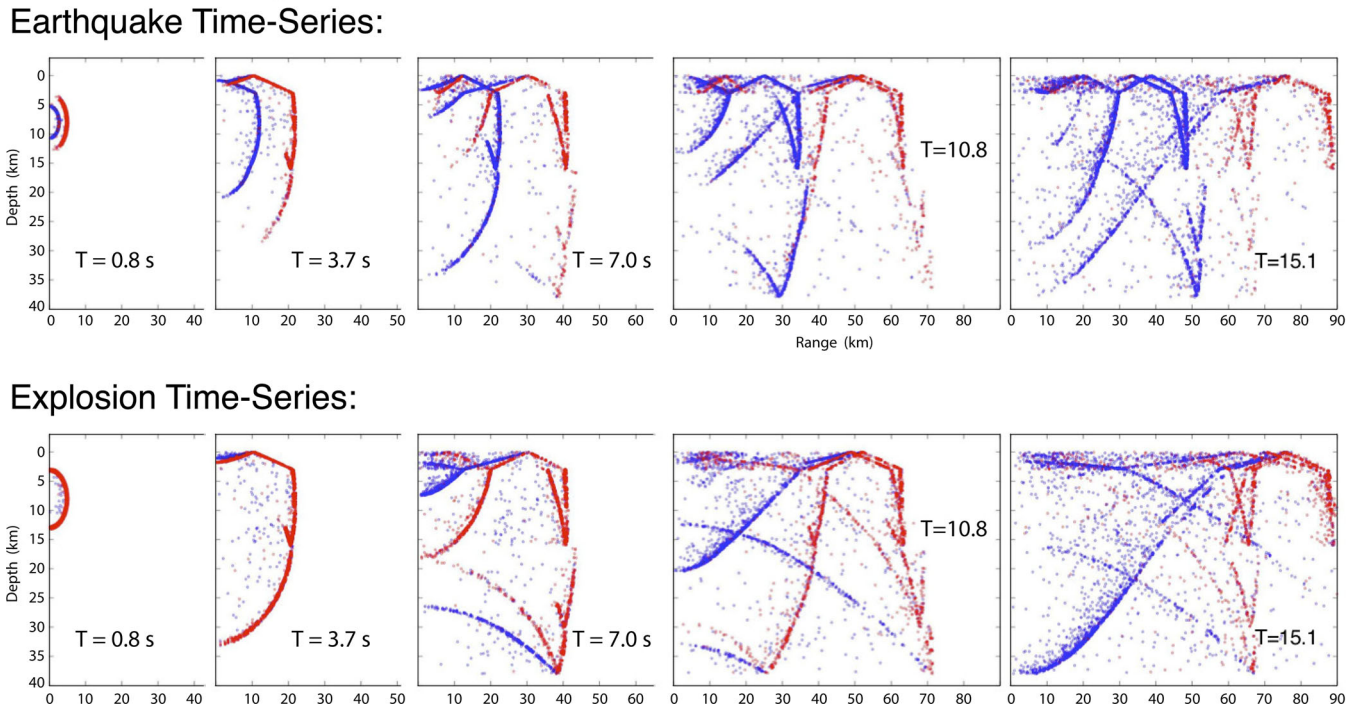


Figure 3. Energy propagation visualized as a time-series plot of phonon locations in a model cross-section. Note the initial absence of the *S* wave fronts (blue) in the explosion time-series and its progressive development with time due to multiple scattering and interaction with the free surface.

a picture of the energy transport throughout the model begins to emerge. Records of the phonon travel paths can be used either in bulk or via selection criteria to visualize this energy transport. In bulk, the data can be used to produce movies or still-frames of the evolving wave front throughout the 3-D model. Via selection criteria, just those phonons that, for example, interact with the surface within a specified gather radius of a hypothetical seismometer can be collected and used to produce seismic waveforms of surface movement at the given location. With a sufficient quantity of virtual seismometers, traveltime curves can also be produced.

3.2 Visualizing output

RADIATIVE3D produces output suitable for visualization with external tools. The output capabilities fall into two major categories: (1) event reporting, and (2) seismic energy binning. Event reporting means reporting the progress of individual phonons in a play-by-play manner. Examples of events in this context include: generation, scattering, crossing a model boundary, free-surface or discontinuity interface reflection and transmission, etc., or simply reporting the position of each phonon at regular time intervals. When these events are analysed in post-processing, detailed pictures of energy propagation can be constructed. One of the first visualization tools developed was a GNU Octave script to produce videos illustrating *P* and *S* energy propagation in the Earth model.

3.2.1 Wave front and coda evolution

Fig. 3 shows energy propagation visualized as a time series plot of phonon locations in a model cross-section. Individual wave fronts, and the coda developing behind them, can be easily discerned in these plots. Red dots represent *P* phonons, blue dots represent *S* phonons. Reflection and refraction occurs at interfaces between model layers at which velocities are discontinuous. Conversions between *P* and *S* polarization are driven by both scattering and reflection/transmission events. A sequence of still frames showing small increments of simulation time can be assembled into videos showing wave front and coda evolution. Producing event logs with sufficient detail to show clear wave-front evolution requires on the order of a few thousands or a few tens of thousands of phonons, and are produced in approximately ten minutes of CPU time on Intel i7 or comparable workstations.

3.2.2 Seismometer output

The other form of output which can be analysed for quantification or visualization is binned seismic energy. RADIATIVE3D allows for the placement of virtual seismometers along any surface designated as a collection surface. Whenever a phonon interacts with a collection surface within a specified gather radius of a seismometer, the energy of that phonon is recorded by the seismometer, decomposed into three Cartesian energy axes, and binned into time windows. At the end of simulation, these energy bins are output and can be interpreted as seismic energy traces, suitable for making envelope plots. Fig. 4 shows seismic envelope plots produced by the RADIATIVE3D code. The traces represent amplitude of the 2.0 Hz signal arriving at a virtual seismometer approximately 800 km from a hypothesized earthquake source (left) and explosion source (right) synthesized in a 3-D model of the Lop Nor, China region having a simple deterministic structure and trial statistical structure. The quantity presented as 'amplitude' is the square root of accumulated energy flux in each time bin, having been decomposed into directional components based on the direction of particle motion of the incident ray, accounting for polarization mode and angle. Accumulated energy flux is the sum of the path attenuated energy carried by phonons interacting with the gather surface within the time bin divided by the gather area and the time bin width. Each phonon is

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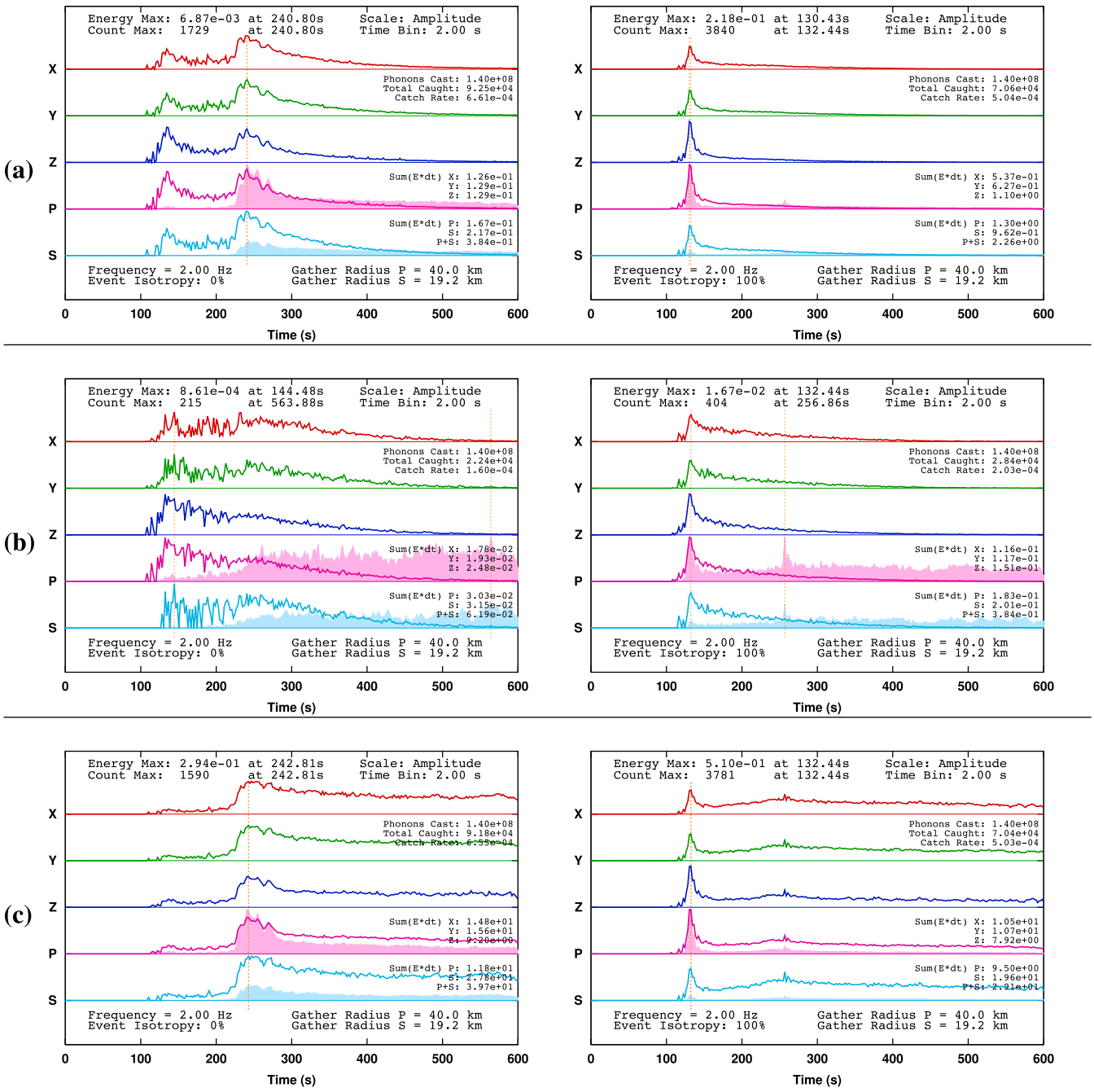


Figure 4. Envelopes produced by RADIATIVE3D for a shallow-focus earthquake (left) and explosion (right) as recorded by a virtual seismometer at 799 km range along the Lop Nor to MAK path in our Lop Nor model. Traces are shown for components of motion NS (Y), EW (X), and vertical (Z). Additional channels P and S denote instantaneous polarization mode of phonons arriving at the receiver. Trace amplitudes are proportional to the square root of the per-area per-time energy flux rate, decomposed by directional channel (X, Y, Z) or by polarization state (P, S). Shaded area under P and S curves represents raw phonon capture count, which neglects intrinsic attenuation effects and is not square-rooted, and serves a primarily diagnostic purpose. Annotations appear on each plot indicating parameters controlling the simulation and statistics describing the output. Vertical dotted red lines give the location in time of the channel-summed energy maximum and the phonon capture maximum, which overlap in some envelopes. The maxima values and time coordinates are also listed at the top of each figure. The bottom left corner of the figure includes the frequency of the run (in this case 2 Hz) and the isotropy of the moment tensor representation of the source (0 per cent indicates earthquakes, 100 per cent indicates explosions). Phonon statistics on the right side of the figure include the total number of phonons cast during the simulation (140 million), the total phonons caught at the location, and the catch rate. Also listed on the right side is the sum of the energy captured, under the assumption that each phonon departs the event source with unit energy. The total energy across all channels is indicated as $P + S$. The time bin width in the above simulations is 2.0 seconds, indicated in the upper right corner, or 4 cycles at 2.0 Hz. Row (a): reference synthetics chosen as a baseline of comparison for experiments varying heterogeneity parameters. (See Section 4.) Row (b): velocity perturbation ϵ increased from 4 per cent to 8 per cent. Row (c): synthetics with effects in intrinsic attenuation disabled.

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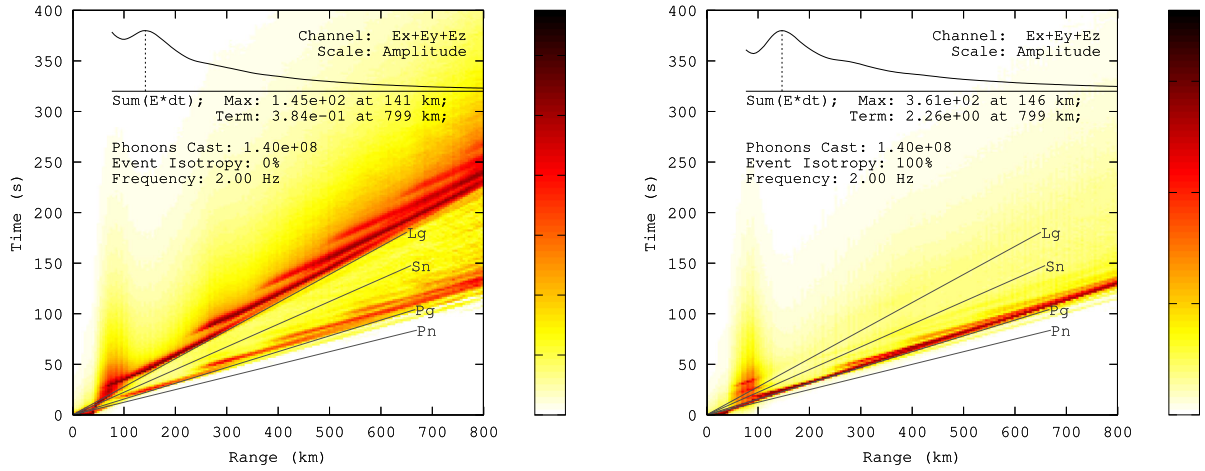


Figure 5. Traveltime curves produced by RADIATIVE3D for shallow focus earthquake (left) and explosion (right), corresponding to the envelopes in row (a) of Fig. 4. Image density is proportional to the square root of energy flux arriving on all three axes, thus depicting an amplitude-like signal. A range-varying normalization curve is applied to the amplitudes so that the seismic phases remain distinct at all ranges. The normalization point is the midpoint between the average value and the peak value of the envelope at each distance range, and provides for good visual separation of phases. An overlay plot shows the time-integrated energy flux plotted on a square-root scale against range to show the overall amplitude decay with distance, omitting the very near range. Annotations identify the maximum of this integrated energy value and the value at the terminal range. Additionally, as with the envelope plots, the number of phonons cast, the frequency of the simulation, and isotropy of the event source are noted. Phase markers denoting the group velocities of the four regional phases are also shown for reference.

assumed to depart from the event source with unit energy. Attenuation based on scattering or geometric spreading need not be considered, as both these factors already control the likelihood of interaction with the gather surface. A unique feature of the phonon method is that the instantaneous polarization state of the phonon is known at the time of interaction with the surface, and this allows us to present two additional signal channels showing the decomposition of energy based on instantaneous *P* or *S* polarization state. Note that the polarization state decomposition does not cleanly separate the *Pg* and *Lg* arrivals. This is due to mode conversion occurring at the interface of the crust with a thin sediments layer at the surface of the model. The *P* and *S* amplitude channels denote the instantaneous state, not the propagation mode of the bulk of the phonon's travel history. We set the depth of the explosion event to 2.0 km below the surface, and the depth of the earthquake event to 6.0 km below the surface, with a focal mechanism of the 2003 March 13 earthquake given by Selby *et al.* (2005).

Traces from multiple seismometers arranged in a linear array can be combined to make traveltime curves. Fig. 5 shows the traveltime curves from the earthquake and explosion simulations that were presented in Fig. 4, row (a). Each curve combines output from 160 virtual seismometers linearly spaced from the source to the maximum range at 800 km. Image density is proportional to the amplitude signal described above. A range-varying normalization curve is applied to the amplitudes so that the seismic phases remain distinct at all ranges. Thus, the image density is not proportional to absolute amplitude, but is relative to the normalization point. These plots allow the identification of seismic phases based on group velocity and time offset.

Discretization of take-off angle (TOA) for phonons generated at the event source implies that care must be taken in choosing an appropriate gather radius for seismometers at long range. In the absence of the randomizing effect of scattering, coarsely-chosen angular increments can result in unreachable zones at range. The gather radius should be chosen large enough to sufficiently average a surface area that can be reached by at least a small number of TOAs, while still maintaining the desired degree of localization in the signal output. Increasing the TOA density can also fulfil this purpose, but comes at a performance cost of searching larger arrays when probabilistically choosing the TOA for each new phonon, as the directionality of the event source is treated through weighting the probability of each discrete angle. An appropriate balance between TOA density and gather radius will allow stable signal output with a manageable number of phonons sprayed from the event source. For the frequency bands and ranges presented here, stability (where increasing the number of phonons sprayed does not appreciably change the shape of the simulated coda) was achieved with 140 million phonons per simulation, with gather radii of 20 to 40 km at 800 km range. TOAs were discretized into approximately 5.2 million discrete angles representing nearly equal solid-angular increments chosen via a triangular tessellation procedure. Single-threaded computation time on Intel i7 and comparable desktop workstations in our lab took between 4 and 7 hr per simulation. Earthquake simulations generally take longer than explosion simulations, due to the greatly increased rate of scattering that occurs in the more strongly excited Lg wave front.

4 EXAMPLE APPLICATION

4.1 Lop Nor region: deterministic model and data

We focused our modelling efforts on the Lop Nor, China region (Fig. 6) because it is one of few regions in which nuclear tests and earthquake waveforms are well recorded at accessible global seismic stations and networks. Sykes & Nettles (2009) found that more than half of the earthquakes in the Reviewed Event Bulletin (REB) of the International Monitoring System (IMS) occurred within 100 km of six Lop Nor test sites from 2000 through 2008. Los Alamos National Lab (LANL) has also constructed maps of regional wave propagation efficiencies in this region (Phillips *et al.* 2001). Lop Nor is located near the southeastern side of the Tian Shan, a region of moderate earthquake activity and contemporary horizontal compressive stress in the earth's crust, which can be an important factor in inducing tectonic release from underground nuclear tests in this region.

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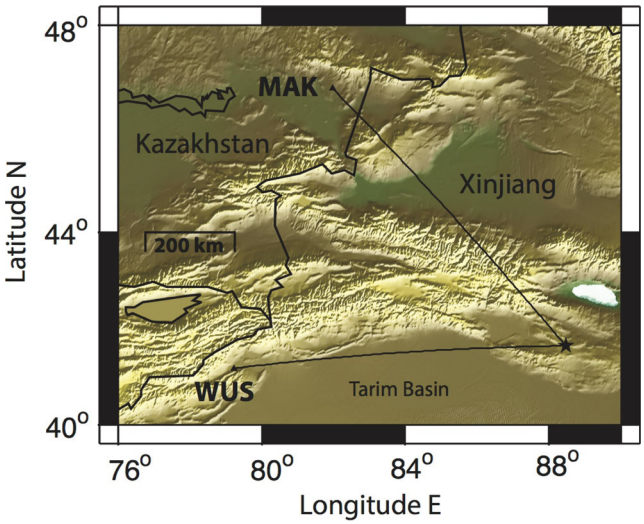


Figure 6. Lop Nor test site and paths to regional seismic stations MAK and WUS on which we have concentrated modelling tests.

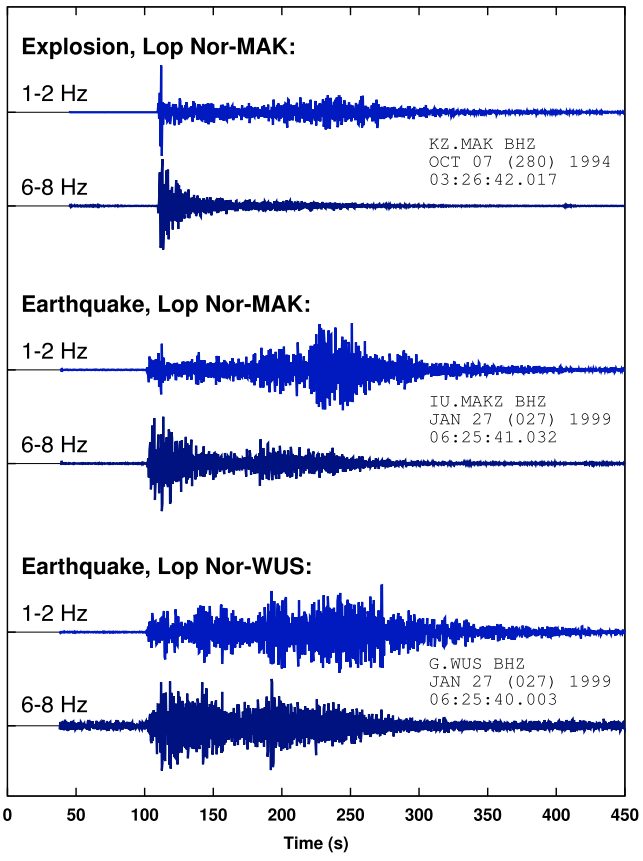


Figure 7. Vertical component band-passed data recorded at stations WUS and MAK in the Lop Nor region.

The Chinese station WMQ (Urumqi) is about 250 km (2.15°) from Lop Nor. Stations MAK and WUS are about 760 km (6.85°) from Lop Nor. For the earthquake and explosion data we use seismograms from events downloaded from IRIS-DMC.

Fig. 7 shows some waveform examples from the Lop Nor region. The data come from an explosion at the Lop Nor site as recorded at station MAK at great circle distance 6.85° and an earthquake in the same area recorded at both MAK and WUS. WUS is approximately at the same distance from Lop Nor as MAK but at a different azimuth. Fig. 7 shows data band passed between 1–2 Hz and between 6–8 Hz for each recording. Not only are there differences between the earthquake and explosion data at MAK, there are also differences between the earthquake traces at MAK and WUS in the time windows commonly used to measure *Pn/Lg* and *Pg/Lg* amplitude ratios.

Using known elevations of Lop Nor and seismic stations MAK and WUS, along with Moho depths from the Cornell Moho model (http://atlas.geo.cornell.edu/geoid/imagegrid.html) and layer profiles from CRUST 2.0 (Laske *et al.* 2011) at those same locations, we located and oriented a set of five crust layers (sediments, upper, middle, and lower crust, and top layer of the mantle). Additionally, we defined an additional 16 mantle layers from AK-135-F (Kennett *et al.* 1995) and regionalized perturbations (Gudmundsson & Sambridge 1998) to a depth of 859 km. These layers served as an initial deterministic background model (Fig. 8) in which to run synthetics. Although our software is capable of simulating laterally varying models that include velocity gradients defined on a tetrahedral mesh, we used only models consisting of layers of uniform background and statistical parameters in the present work. Gradient-like structures were approximated as a succession of layers varying in stair-step fashion. In models of this construction, phonon trajectories are straight lines, and scattering mean free paths are spatially stationary within individual layers. Reflection and refraction occur at interfaces between layers.

4.2 Statistical model

To determine our statistical model for the Lop Nor region, we first performed a series of experiments to examine the sensitivity of regional phases to the shape of the heterogeneity spectrum. Results of these experiments are displayed as coda envelopes by component of motion and by the sum of the relative contributing phonon states (*P* or *S*) at receiver (Fig. 4). Spectral shape parameters were varied for a von Kármán spectrum. The advantage of this spectrum is that it can incorporate a more realistic medium behaviour over a broad band of wavenumbers, reproducing a fractal-type behaviour beyond a corner wavenumber (Margerin & Nolet 2003). The parameters varied are: the fractional fluctuation ϵ of seismic wave velocities; a parameter ν controlling the density perturbation as a factor multiplying the *P* velocity fluctuation; a scale length a that controls the wave number corner; and a von Kármán Hurst order number κ that controls the slope of the power spectrum after its corner. Following Sato *et al.* (2012), who simplified formulas for scattering radiation patterns by assuming ϵ is identical for *P* and *S* waves, we take $\epsilon = \frac{\Delta V_P}{V_P} = \frac{\Delta V_S}{V_S}$, equivalent to assuming $d \ln V_S / d \ln V_P = 1$. Experimental verifications of this assumption are few. From a local array above the Tonga subduction zone, Koper *et al.* (1999) estimated $d \ln V_S / d \ln V_P$ to be in the range 1.1–1.5 at scale lengths on the order of 10–100 km. In the lower mantle at larger scale lengths (1000 km), $d \ln V_S / d \ln V_P$ has been estimated to be >2.5 (Romanowicz 2001).

The sensitivity of regional phase codas to heterogeneity parameters can be understood by examining the behaviour of the mean free path (MFP) and the dipole projection (DP), while varying a single parameter and holding the others constant. The MFP and DP calculations provide an easy way to understand the general effects of the parameters. The MFP reports how far a phonon will travel, on average, before it encounters a scattering event. Longer MFPs mean less scattering. The dipole projection is simply the mean cosine of the angular deflection of a phonon path after a scattering event. It varies from −1 to 1. A value of 1 means it is 100 per cent likely

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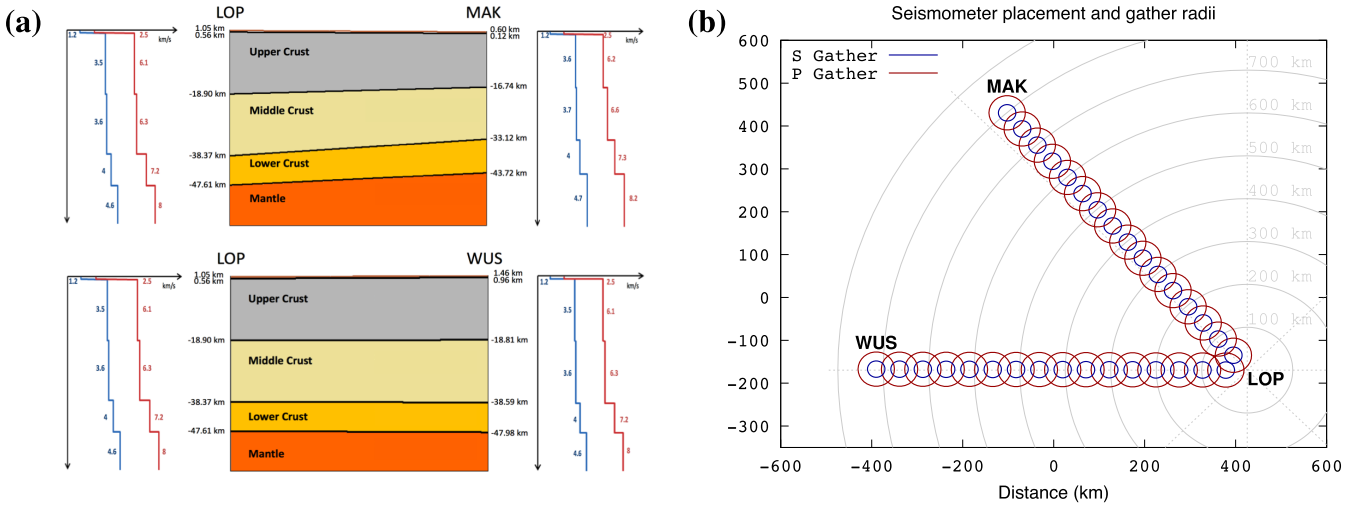


Figure 8. (a) Simplified Lop Nor Earth model composed of layers of uniform background velocity separated by planar tilted interfaces oriented to match depth profiles at Lop Nor, station MAK, and station WUS. A thin sediment layer (0.5 km) is included and sits on top of three crust layers extending to a depth of 45 km, and 16 mantle layers down to a depth of 859 km. (b) Map view showing relative locations of Lop Nor, MAK, and WUS in model coordinates and two virtual seismometer arrays represented by gather-radius circles. Each array consists of 160 evenly spaced seismometers of which every tenth one is shown. Results presented here will focus on the LOP to MAK path.

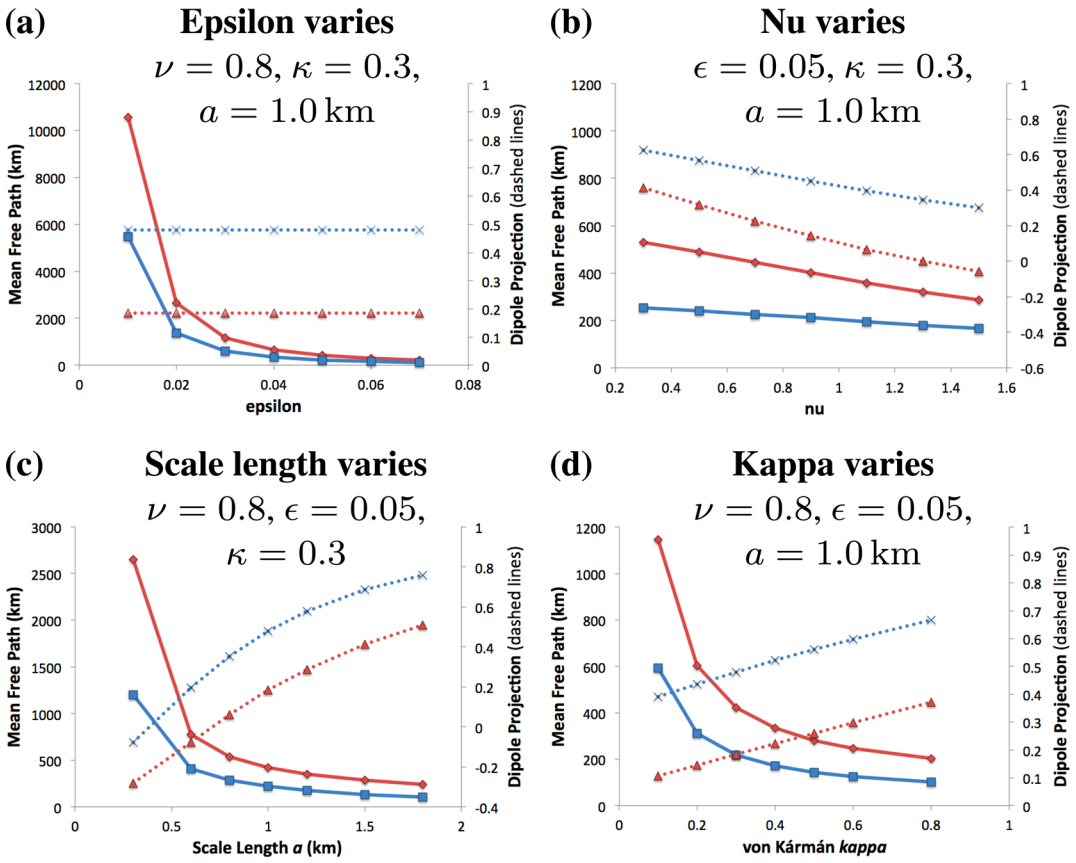


Figure 9. Effects of heterogeneity parameters ν, ϵ, a and κ, on mean free path MFP (average distance between scattering events) and dipole projection DP (average cosine of deflection angle) for 1.0 Hz propagation in a background medium having P and S velocities $v_P = 6.93\,\mathrm{km\,s^{-1}}$ and $v_S = 4.00\,\mathrm{km\,s^{-1}}$. The DP value serves as a measure of scattering directionality, with positive values indicating dominant forward scattering, negative values dominant back-scattering. Fixed parameter values are selected in the neighbourhood of values used for the studies in this paper. Blue series indicates trends for S propagation and red for P propagation.

to continue in its forward direction, and a value of -1 means it is 100 per cent likely to be totally scattered backwards. Intermediate values indicate wide deflection angles and/or a balance of forward and back-scattering.

In Fig. 9, values for MFP and DP were computed at 1.0 Hz in a background P and S velocity appropriate for Earth's crust, and serve to illustrate the changes in scattering behaviour that can be expected from variation of individual statistical parameters. The

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Table 1. Reference heterogeneity parameters used for synthetic comparisons in the simplified Lop Nor Earth model.

	ν	ϵ	a	κ	Q_S
Sediments	0.8	0.01	0.25 km	0.2	50
Crust	0.8	0.04	0.2 km	0.3	1000
Mantle	0.8	0.008	0.2 km	0.5	300

MFP effects are read on the left side y-axis as solid lines, and the effects on DP are read on the right side y-axis as dashed lines. The parameter being varied is plotted on the x-axis. The red lines show the effect of scattering on the P-wave phonons, and the blue lines show S-wave phonons.

In the subsections that follow, synthetic envelopes for earthquake and explosion sources are computed in media similar to that of Fig. 9, although at a higher frequency of 2.0 Hz, and presented to assess the impact of each individual parameter. Comparisons are made to a common reference synthetic, shown in row (a) of Fig 4, with parameters given in Table 1. Q_P is assumed to greater than Q_S by a factor of approximately $\frac{9}{4}$, consistent with an assumption of viscoelastic attenuation purely in shear and a Poisson's ratio of $\frac{1}{4}$ (Anderson 1989).

4.2.1 *Magnitude of velocity fluctuation* ϵ

The dashed lines in Fig. 9(a) reflect a constant S-wave dipole projection of 0.48 and a constant P-wave dipole projection of 0.18. For sufficiently small fluctuations and small scale length, varying ϵ has essentially no effect on the scattered phonon's preferred direction of travel after scattering because the magnitude of ϵ does not affect the shape of the scattering radiation pattern at a scattering event. On the other hand, there is an inverse quadratic decrease in the MFP for both P and S phonons for this range of increasing velocity perturbation. We should expect to see increased coda production for both S and P waves in the synthetic envelopes with increased ϵ.

Fig. 10 compares synthetic z-component envelopes at 2.0 Hz for crustal ϵ values of 0 per cent, 4 per cent, and 8 per cent for an earthquake simulation and 4 per cent and 8 per cent for an explosion simulation. Fig. 4(b) shows envelopes for all components at $\epsilon =$ 8 per cent for the earthquake and explosion simulations, which can be compared against row (a) containing the $\epsilon = 4$ per cent reference simulations. For the reference earthquake, the maximum energy and maximum phonon count in the reference occur at the time of the Lg phase (241 s). For the reference explosion, the maximum energy and maximum phonon count in the reference occur within the P coda (130 s). We see a number of effects in the seismic envelopes caused by the change in the heterogeneity parameter, including change in total energy and phonons arriving at the receiver, change in maximum energy, onset of P and Lg phases, shape of the direct phase arrival, and the coda following the direct arrival.

The combined results of the MFP/DP plots and synthetic envelopes show that with higher velocity perturbations, the phonons have smaller mean free paths, thus encountering more scattering events. As velocity fluctuations increase, the arrival time of the maximum energy shifts from the Lg arrival window to the P arrival window.

The scattering caused by the increase in the velocity perturbation produces a marked redistribution in energy, stretching it out over a longer period of time. This is represented by a slower decay of coda envelopes, the broadening of the pulses, and a pronounced delay in the phonon arrivals (shaded areas in the P and S channels, Fig. 4). These results reflect the increased production of scattered

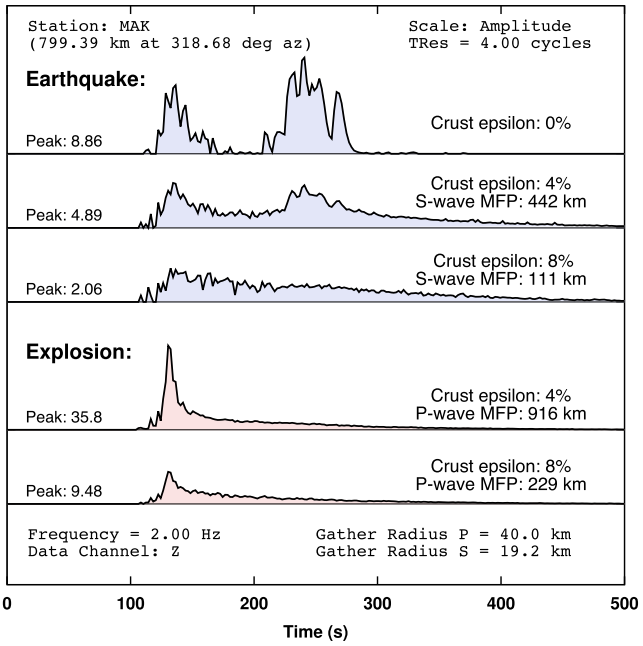


Figure 10. Effects of velocity perturbation ϵ on earthquake and explosion z-component coda envelopes. The peak amplitude value is noted for each trace, as is the ϵ value used in the crust layers and the resulting mean free paths in the middle crust layer for each case.

P and S energy as well as conversions between and P and S energy, particularly for the explosion. Without scattering, almost all of the explosion energy is in the P arrival. With the scattering, in addition to coda energy following the P-arrival, there is significant energy arriving at the time we would expect to see Lg. This delayed energy, however, is reflected in a slower coda envelope decay following the P phases, which cannot easily be mistaken for the typical Lg coda envelope observed from earthquakes.

4.2.2 *Magnitude of density perturbation* ν

The density perturbation ν is a multiplier of the velocity perturbation: $\frac{\Delta\rho}{\rho} = \nu\frac{\Delta V_P}{V_P}$. Fig. 9(b) shows the effects of density perturbations from 0.3 to 1.5 on MFP and DP. Shearer & Earle (2008) take $\nu = 0.8$ for small-scale heterogeneity (1–10 km). For large-scale (>1000 km) structure in the mantle, geodynamicists take ν on the order of 0.2 or less because large values cannot be sustained by buoyancy (Forte & Woodward 1997).

Higher values of ν increase the backscatter fraction while only moderately affecting the forward scatter fraction of the basic scattering radiation patterns. The effect of the heterogeneity spectrum is to modulate the basic scattering patterns and can strongly inhibit the backwards scattering fraction in the wavenumber regime $ka \gtrsim 1$. At the wavelength and heterogeneity scales studied in Fig. 9 and in the envelope synthetics, this inhibition of the backwards scattering fraction moderates the relative significance of the increase due to increasing ν, thus resulting in a more gradual decrease in mean free path compared to the effects of the other parameters. The effect on the balance between forward and backward scattering is seen in the decreasing dipole projection. The decrease in MFP implies an increase in the overall rate of scattering, and although this increase is only slight, the fact that the increase comes mainly in the backward scattering direction means that the scattering that does occur efficiently removes energy from the wave front and/or shifts

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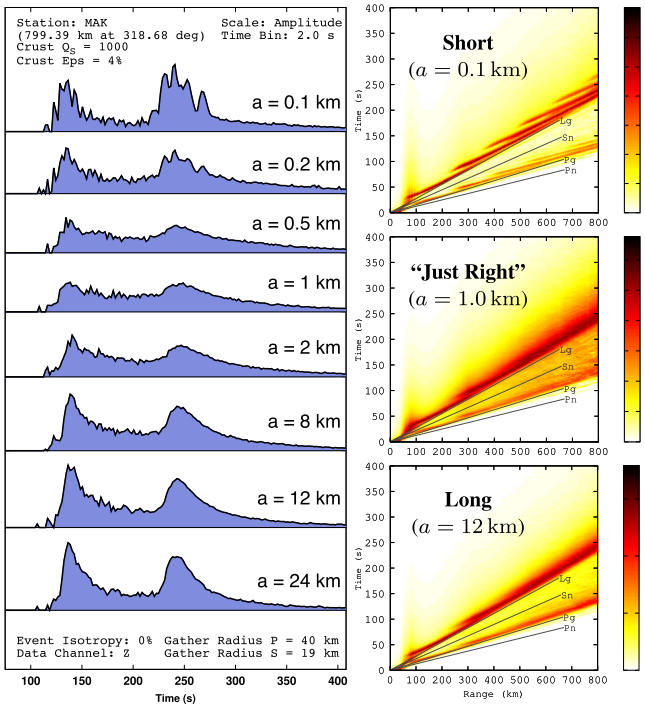


Figure 11. Simulated effects on earthquake seismograms of a von Kármán heterogeneity spectrum with varied scale length a applied in the crust layers (from just below sediment layer down to 45 km depth). Note that the coda of individual multiple Moho-reflected S waves comprising Lg overlap as the scale length increases, and that coda production is optimized when the scale length is on the order of the wavelength.

it into later time windows. Thus ν variations can be important in controlling the attenuation of peak amplitudes and the broadening of coda for regional phases, even in the parameter ranges studied here.

We produced synthetic envelopes for a doubling of ν from 0.8 to 1.6 against our reference heterogeneity parameters. The effects of this doubling on coda envelopes were much less pronounced than the effects observed from our ϵ variations; hence, we do not show synthetic envelopes. Maximum energy occurs at the same time as the reference, and the maximum phonon count occurs only slightly later (255 s). The P onset is sharper and of lower amplitude, but the effects on coda envelopes of plausible density perturbations are much weaker than those of plausible velocity perturbations. In general, the influence of density variations is to produce more scattered energy into the back half space, but as the effect of the heterogeneity spectrum can be to reduce this backscatter fraction, the overall impact of ν variations can be slight in some cases. Nonetheless, the ability to adjust ν could be important in other seismic coda studies, such as those involving partial melt (Hong *et al.* 2004).

4.2.3 Scale length $\boldsymbol{a}$

Fig. 9(c) shows the effects of scale length variations from 0.5 to 12 km on MFP and DP. For the velocities and frequency (2 Hz) used in coda envelope simulations, the wavelength in the crust is 2 km for S waves and 3.4 km for P waves.

Fig. 11 shows synthetic envelopes for a range of scale lengths and traveltime curves for a subset of lengths from 0.1 km to 12 km. These values were prescribed in the crust layers, and the results allowed us to examine the effects of the scale length being greater than or smaller than the wavelength. The mean free path decreases as the scale length increases, but the dipole projections asymptotically approach forward scattering values. Not surprisingly, the competition between these two effects results in the strongest coda production when the scale length is on the order of the wavelength. When the scale length is much smaller, there are too few scattering events to significantly affect waveform shape. When the scale length is much longer than the wavelength, scattering events become frequent, but are non-deflectionary, and cause only minimal time lags with reference to the un-scattered trajectories, resulting primarily in a smoothing of the waveform but not significant coda generation.

4.2.4 Hurst parameter $\boldsymbol{\kappa}$

The magnitude of the fluctuation power spectrum for a von Kármán medium is flat with wavenumber in the domain $ka \ll 1$, and decays as a negative power of wavenumber in the domain $ka \gg 1$. The Hurst parameter κ controls the decay rate at scales smaller than the corner scale a, in the $ka \gg 1$ domain. The von Kármán medium is characterized by heterogeneities that are 'rougher' at small length scales than the Gaussian medium, in which the power spectrum decays much more rapidly in the $ka \gg 1$ domain (Frankel & Clayton 1986; Sato *et al.* 2012).

Fig. 9(d) shows the effects of κ variations from 0.1 to 0.8 on MFP and DP. As κ increases, the MFP decreases, which causes scattering delays in the energy arrival. As the slope of the heterogeneity spectrum flattens below the corner wavenumber, the scattering is increasingly in the forward direction even though scattering events are more frequent. The effect of increased scattering at higher κ will have less effect on redistribution of energy and stretching of coda envelopes, but may have a stronger effect on equilibrating energy on the three components of motion due to a more frequent sampling of heterogeneity radiation patterns that control polarization and wave type (P or S). The heterogeneity power, however, will decrease faster at wave numbers above the corner wave number for higher κ. We found the effect of κ variation to be minor in our experiments at distances beyond several hundred kilometres and at frequencies between 2 to 4 Hz. For our assumed frequencies and scale lengths, however, $ka \sim 1$. In contrast, for $ka \gg 1$, Saito *et al.* (2002) and Sato & Fehler (2016) have found that variations in κ control envelope broadening. Since visible effects of κ are not easily discernible in the distance ranges and frequencies we examined, we do not show synthetics for this experiment.

4.2.5 Intrinsic Q effects

RADIATIVE3D allows us to treat intrinsic viscoelastic attenuation separately from scattering to help determine their relative contributions to the total attenuation of peak coda amplitudes or to separate their contributions to measured coda-Q's. Fig. 4(c) shows coda envelopes modelled in the absence of intrinsic attenuation, and can be compared against the reference envelopes in Fig. 4(a) that included Q values as specified in Table 1. For the earthquake, the 2 Hz total integrated energy without Q^{-1} included is about 100 times that of the reference envelope, and the peak energy value is about 50 times that of the reference. The phonon capture is about the same. The effect of Q^{-1} is to reduce the amount of energy in each phonon as a function of time travelled, but it does not reduce the number of phonons reaching the target. Perhaps more importantly, the test also shows that not adding intrinsic attenuation results in the coda

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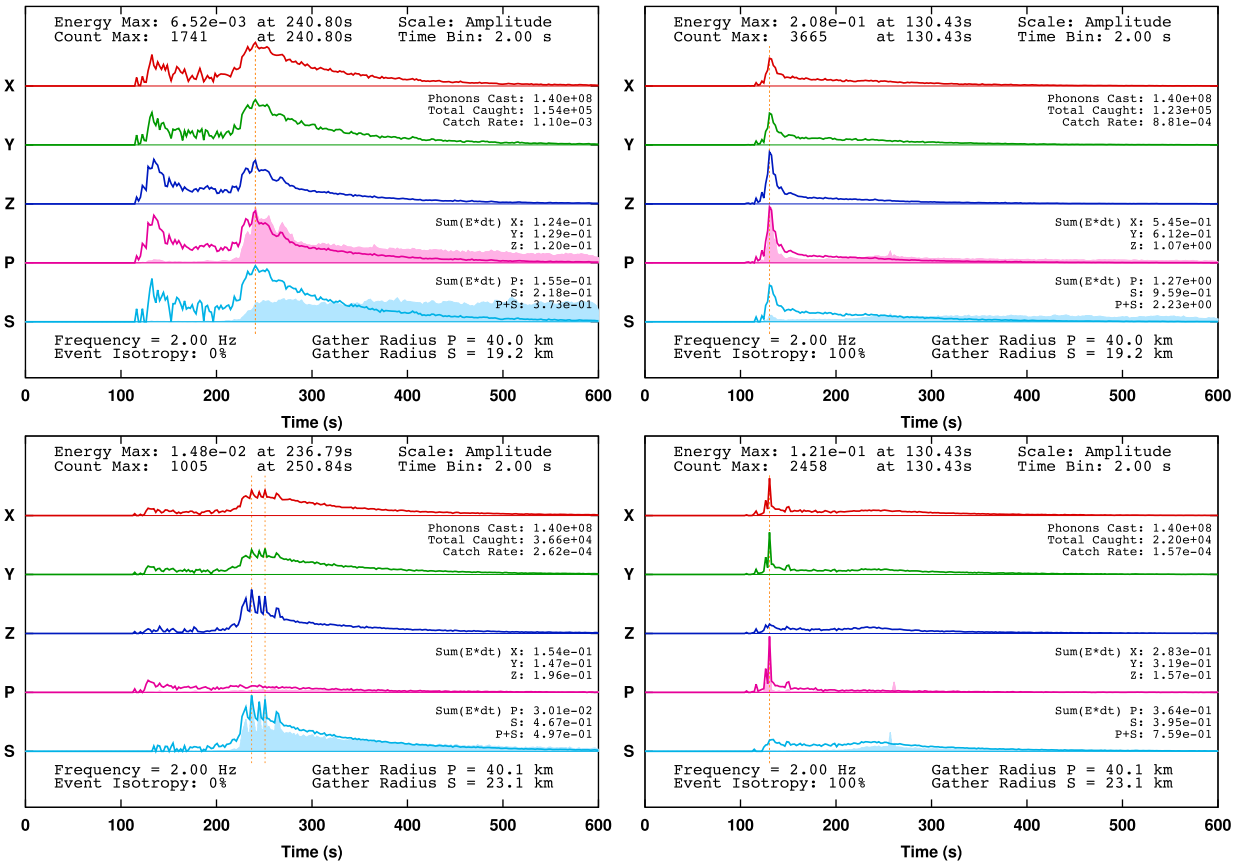


Figure 12. Synthetic envelopes for surface layer effects. Top left: earthquake simulation with sediment layer included; top right: explosion with sediment layer; bottom left: earthquake without sediment layer; bottom right: explosion without sediment layer.

Table 2. Deterministic and statistical properties and resulting mean free paths of sediment and upper crust layers in the models exploring effects of the sediment layer on coda shape (Fig. 12). Heterogeneity parameters ν, a and κ are unchanged from those listed in Table 1 for the reference synthetics.

Layer	v_P (km s^{-1})	v_S (km s^{-1})	density (g cm^{-3})	Q_P	Q_S	ϵ	MFP_P (km)	MFP_S (km)
Sediments	2.50	1.20	2.10	163	50	0.04	114	39
Upper Crust	6.13	3.53	2.75	2262	1000	0.04	826	411

energy being unrealistically extended in time. Accurate models of intrinsic Q in the crust and just beneath the Moho are essential for proper modelling of Sn and *Lg*. In future work we will examine trade-offs between scattering parameters and intrinsic attenuation and test the multiple lapse-time window analysis method (Fehler *et al.* 1992; Hoshiba 1993) for separating intrinsic attenuation from apparent scattering attenuation.

An interesting artefact appearing in the explosion coda of Fig. 4(c) is a small spike at about 255 s. On the basis of its measured slowness, traveltime, and polarization, we believe this spike originates from *P* to *SV* scattered phonons near the source that are then reflected from the 410 discontinuity. This S410S wave is a small-amplitude, pre-critical reflection at this range, but its amplitude increases with range beyond 500 km corresponding to incidence angles at which the *SV* to *P* transmitted wave is no longer excited beneath the 410 discontinuity. These pre-critically reflected phonons would normally be sufficiently attenuated so as not to contribute to the resulting envelope. In the absence of attenuation these phonons can make visible smaller amplitude phases normally lost in the scattered coda.

4.3 Surface layer effects

A highly heterogeneous, strongly scattering, sediment layer may affect regional phase coda (Baumgardt 2001), particularly for shallow emplaced explosions. Since detailed information on the velocities, thickness, and lateral variation of the sediment layer is not easily available for this region, we conducted some experiments on its possible effects. Fig. 12 shows synthetic envelopes exploring the effects of a 0.5 km thick sedimentary layer with deterministic and statistical properties given in Table 2. The first row of Fig. 12 shows synthetics very similar to the reference synthetics of Fig. 4(a), and includes the sedimentary layer. The only change from the reference model was an increase in sediment layer velocity perturbation ϵ from 1 per cent to 4 per cent, and a corresponding reduction in scattering MFP in this layer from 1818 km to 114 km for *P* waves and 627 km to 39 km for *S* waves. Although this is a large increase in the rate of scattering in this layer, the layer is thin and the effects on coda shape are subtle. In the second row of coda envelopes in Fig. 12, the sediment layer has been inactivated by replacing its properties with those of the upper crust layer immediately below it, such that the upper crust in this model effectively extends to the

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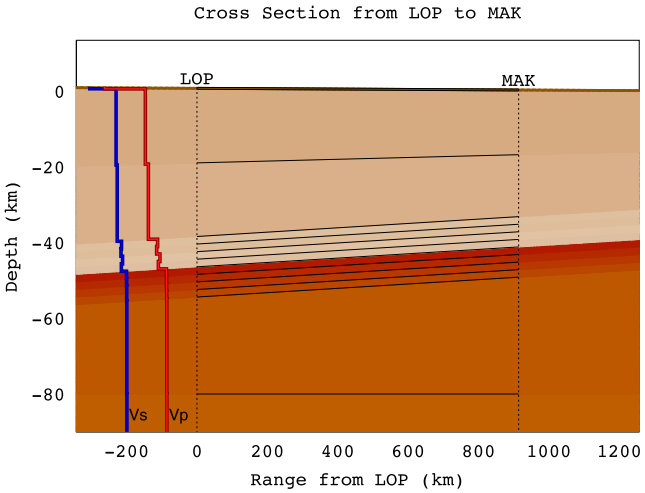


Figure 13. Earth model having a Moho transition region at the base of the crust and a very slight stair-step ramp at the top of the mantle for a path from a southern Xinjiang earthquake to station MAK. Depth dependences of *P* and *S* velocities are also shown. For simulations that included scattering, ϵ was 4 per cent in the crust and 0.8 per cent in the transition region and mantle. Simulations performed in this model had an earth-flattening transformation applied on input to the software to enhance effects of the stair-step gradients and mimic effects of Earth curvature. There is a thin (0.5 km) low velocity sedimentary layer at the top of the model.

surface. This eliminates both the additional scattering in this layer and the refraction, reflection, and P/S conversions that would occur at the crust–sediment interface.

For both the earthquake and the explosion, the most obvious effect of the low velocity sediment layer is on the P/Lg amplitude ratio. Although the differences between earthquake and explosion envelopes are strong in both cases, the effect of the sediment layer will be to slightly reduce the effectiveness of the P/Lg discriminant. There is an apparent tendency for the sediment scattering layer to equilibrate energy on all three components of motion (*X*, *Y*, *Z*).

4.4 Pn and Sn

High frequency *Pn* and *Sn* are traditionally thought of as head waves traveling just beneath the Moho discontinuity. Early attempts to model high frequency *Pn* and *Sn* as classical head waves failed because the amplitude of a classical head wave decays with frequency (ω) as $1/\omega$. These phases are now recognized as interference head waves (Červený & Ravindra 1971). Interference head waves have a representation as series of body waves that are multiply reflected along the underside for either a Moho following Earth's curvature or for velocities that increase with depth beneath the Moho. Our initial experiments assumed a flat earth and homogeneous layers, separated by planar tilted interfaces. In these models energy does not return from the mantle except as scattered waves. With a series of stair-step discontinuities, we have simulated the effects of Earth curvature via an earth-flattening transformation and positive velocity gradients with decreasing radius (Fig. 13). These modifications allow *Pn* and *Sn* to return to the surface after being multiply reflected by and scattered beneath the Moho. Table 3 lists the values of the heterogeneity and *Q* parameters used in this model. These are similar to the reference parameters used in the heterogeneity parameter sensitivity study, but with the addition of a transition region at the Moho. Additionally, intrinsic *Q* values were increased in the crust layer and below, as we found this improved match quality.

Table 3. Heterogeneity and *Q* parameters for simulations in models involving a Moho transition region (Figs 13–16).

	ν	ϵ	a	κ	Q_S
Sediments	0.8	0.01	0.25 km	0.2	50
Crust	0.8	0.04	0.2 km	0.3	2000
Transition	0.8	0.008	0.2 km	0.5	4000
Mantle	0.8	0.008	0.2 km	0.5	1200

Envelopes were synthesized for the 2003 March 13 southern Xinjiang earthquake (m_b 4.8) recorded at MAK for this Moho model. Fig. 14 shows envelope and travel time results of this experiment with scattering from the reference statistical model turned on and off. New sharp arrivals preceding the *Pg* and *Lg* phases are visible in the envelope results. Inspection of the traveltime curves confirms that these peaks have phase velocities consistent with *Pn* and *Sn* phases.

Fig. 15 compares the synthetic *z*-component envelope that includes scattering against the recorded earthquake. The approximate arrivals of the four regional phases are marked. There is good agreement in arrival time and relative amplitude for the *Pn* and *Lg* phases. The *Pg* phase is overly sharp in the synthetic but more diffuse in the recording, although peak amplitudes are in good agreement. The *Sn* peak in the synthetic leads the corresponding peak in the recording by about 10 s and under-represents the relative amplitude by about 20 per cent, but is otherwise in good agreement. Lastly, the rate of *Lg* coda decay is too gradual, indicating intrinsic *Q* values may be too high. We believe that better agreement may be achieved by tuning intrinsic *Q* values as well as extending the Moho transition region a bit further into the upper mantle, adding more layer divisions in that region so that the gradient-enhancing effect of the Earth-flattening transformation can more effectively enhance ray-turning potential, thereby increasing complexity in the *Pn*, *Pg* and *Sn* phases that sample or are affected by that region.

4.5 Synthetics compared with data by frequency band

RADIATIVE3D is run at a single frequency and can create simulated seismogram envelopes at any frequency. This allows analysts to interpret the structural or source origin of frequency-specific effects on narrow band passed seismic traces. Although the radiative transport algorithm assumes ray trajectories are independent of frequency, it incorporates the effects of scattering and intrinsic attenuation through functions that are frequency dependent. Additionally, source-spectral effects can be modelled by applying a weight function to single-frequency envelopes and combining these into a broad-band envelope after simulation.

There is now a substantial body of empirical evidence that regional *P/S* ratios provide poor discrimination below some frequency, typically about 2–3 Hz, and useful discrimination at higher frequencies, (e.g. Fisk 2006). Hence, in Fig. 16 we compare computed envelopes against narrow band passed data at 2, 3 and 4 Hz.

The traces presented Fig. 16 show a good general agreement between the synthetics and the data in (1) the apportionment of energy between the *Pg*, *Pn*, *Lg* and *Sn* windows, favouring *Lg* at 2.0 Hz but *Pg* and *Pn* at 4.0 Hz; and (2) the overall amplitude decay from the 2.0 to 4.0 Hz bands. Although our modelled crustal shear *Q* value (2000) is higher than the coda *Q* values (731–951) predicted by Martynov *et al.* (1999) in this frequency band from observations in the nearby Tian Shan region, we point out that our *Q* values are intrinsic *Q* values modelling thermal losses, and that the contribution from scattering will result in a lower effective

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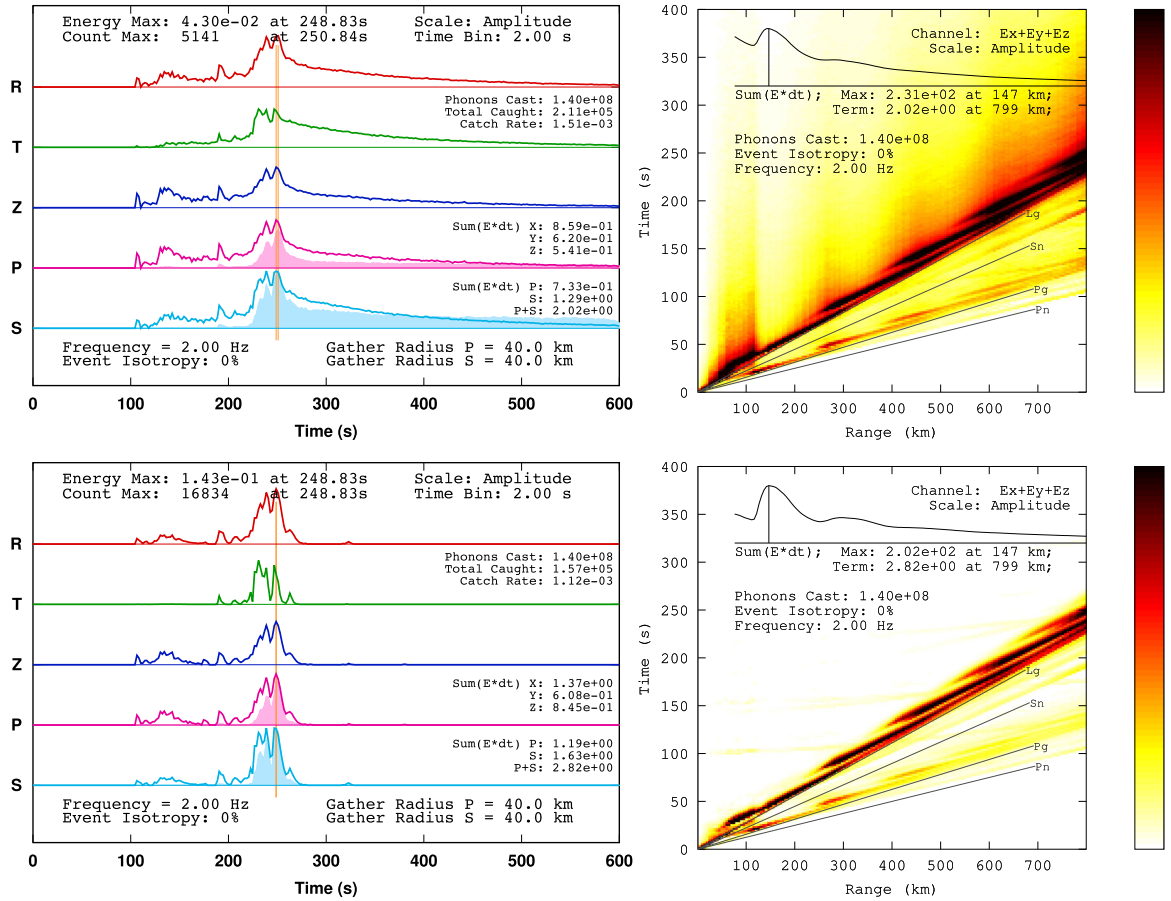


Figure 14. Top: Synthetic envelopes for the Moho transition model (Fig. 13) for the path to station MAK from a focal mechanism patterned after the 2003 March 13 southern Xinjiang earthquake (strike, dip, rake: 125°, 40°, 90°; depth 6 km) with scattering predicted from the heterogeneity parameters in Table 3. Bottom: synthetic envelopes for the same path with scattering turned off.

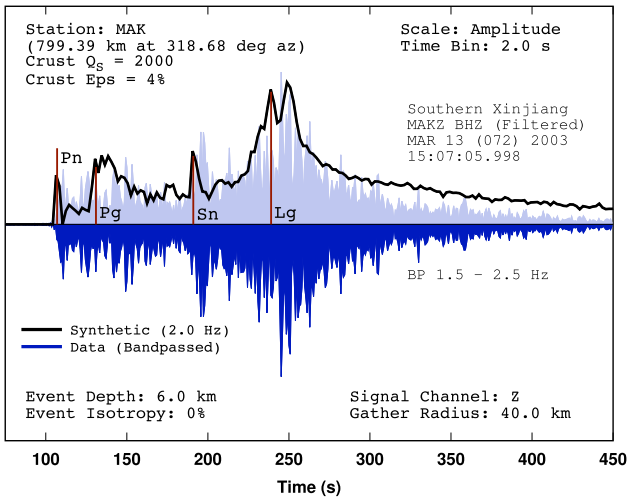


Figure 15. Comparison of filtered recorded data with a 2.0 Hz synthetic coda envelope for a model having a Moho transition layer. Data is band passed from 1.5 to 2.5 H, enveloped, and mirrored across the x axis. The approximate arrivals of the regional phases are marked.

Q value. In fact, we can estimate this effective Q value by assuming it is the inverse sum of intrinsic and scattering Q values, or

$$Q_{\text{eff.}}^{-1} = Q_{\text{int.}}^{-1} + Q_{\text{scat.}}^{-1}$$

with $Q_{\text{scat.}}$ defined for S waves by

$$Q_{\text{scat.}} = \frac{\omega \Lambda_S}{\beta}$$

where Λ_S is the scattering mean free path for S waves (Fehler *et al.* 1992). In our model, the shear mean free path value was 411 km at 2.0 Hz in the crust, resulting in a $Q_{\text{scat.}}$ of 1463 and therefore a $Q_{\text{eff.}}$ of 845, which matches well with the estimates of Martynov *et al.* (1999).

4.6 Tectonic release: regional propagation effects

We investigated the effect on high frequency regional phase propagation of explosion sources associated with components of tectonic stress release. These experiments were inspired in part by the work of Bukchin *et al.* (2001). That study jointly inverted P wave first motions and teleseismic Love and Rayleigh wave spectra for unconstrained moment tensors of earthquakes and nuclear tests in the Lop Nor region. The amount of tectonic release of nuclear tests was expressed as an angle given by the inverse tangent of the ratio of the isotropic to non-isotropic moment tensor components. Our experimental series incorporated moment tensors reported by Bukchin *et al.* for the Lop Nor nuclear tests. Envelopes were synthesized for paths to stations MAK and WUS.

For this series of experiments, we used the moment tensor decompositions of several explosion events published by Bukchin *et al.* (2001) as our starting points. Although they treat the isotropic

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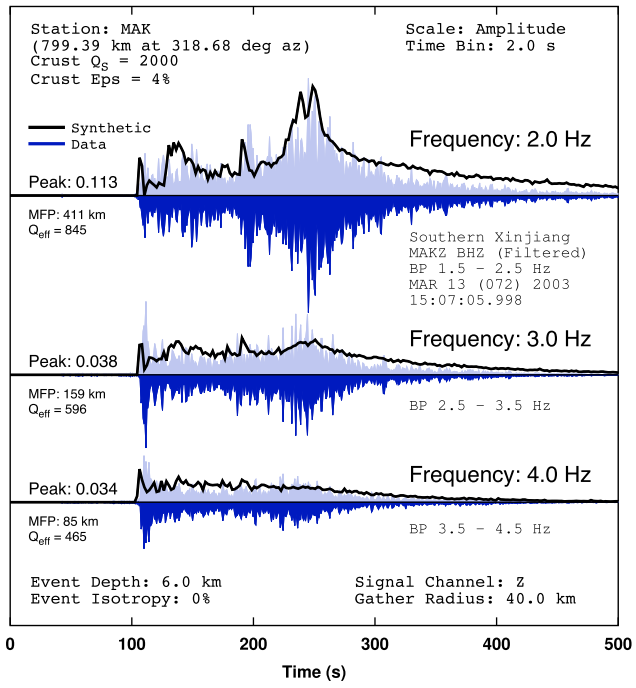


Figure 16. Synthetics compared against data recorded at station MAK for the 2003 March 13 earthquake for frequencies 2, 3 and 4 Hz. Peak amplitudes of the synthetic traces are notated to facilitate comparison between traces. Scattering mean-free paths and effective Q values (combining intrinsic Q and frequency-dependent scattering Q) noted for each trace are for S waves in the crust.

and deviatoric components as vertically separated, in these runs we treated them as co-located and coherent (meaning a single moment tensor specifies the entire source, as opposed to running the deviatoric and isotropic source radiation events separately and summing the synthetics).

Fig. 17 shows an example of one of our event simulations compared to data recorded in several narrow band passed filters for 1 to 4 Hz. We modelled the 1996 June 8 explosion event (m_b 5.8) with 30°, 45° and 60° isotropic angles. We found the best agreement to recorded data at 45°, representing an equal partition between isotropic and deviatoric moments, supporting the findings of Bukchin *et al.* Note the explosion with tectonic release strongly enhances *Lg* at 1 Hz and decreases the *Pg*/Lg ratio in this frequency band, despite being synthesized at an azimuth that favours *Pg* based on the deviatoric component of the moment tensor. This suppression of the *Pg* phase is a result of the isotropic component serving to compensate the tensional principal axis of the composite moment tensor. Although the shift between *Lg* and *Pg* amplitude peaks as frequency increases is qualitatively well matched to the recorded data, there is a slight over-representation of the *Lg* phase, and a marked underrepresentation of the *Pn* phase. The former may indicate the need to further tune crustal Q values, or to more finely scan isotropic angles in the vicinity of 45° to optimize match quality. The latter further illustrates the need to refine model gradient structure in the Moho transition region and upper mantle to enhance *Pn* strength and complexity, as was noted previously.

We finally note that our code does not include fundamental mode surface waves and our tests are for relatively long range (>100 km), and high frequency (>1 Hz). At this frequency and range surface waves are usually strongly scattered away by surface topography and near surface heterogeneity. Numerical simulations by Obermann *et al.* (2013) at ranges on the order of 10 km have found that high

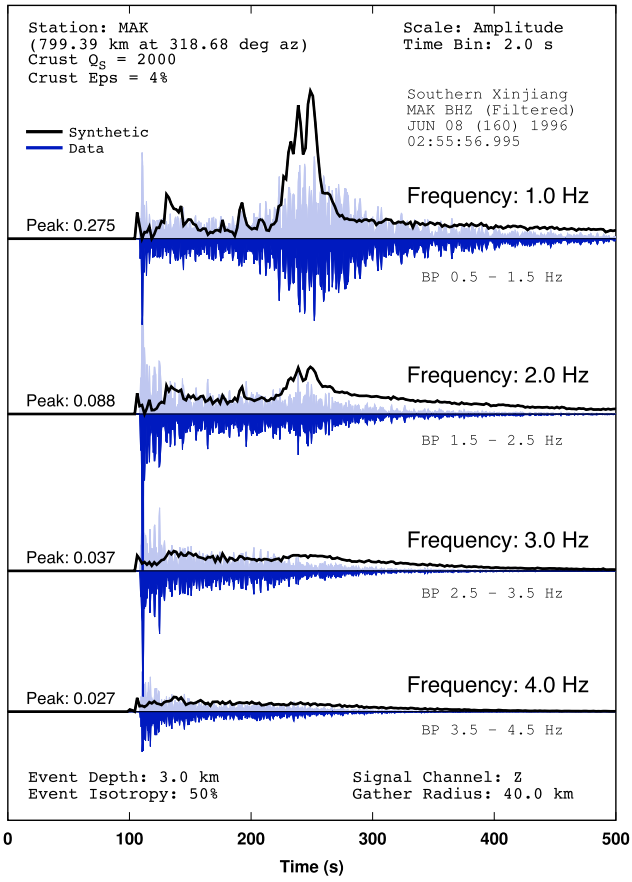


Figure 17. Synthetic envelopes for a Lop Nor nuclear test using source mechanism with tectonic release determined from the surface wave study of Bukchin *et al.* (2001), compared with observed envelopes for the same test. The tectonic release component simulates strike, dip, and rake angles of 280°, 70° 160°. The composite moment tensor models an isotropic angle of 45°, representing an equal contribution from isotropic and deviatoric moments.

frequency surface waves can provide valuable discrimination of source depth and possibly also of tectonic release.

5 CONCLUSIONS

We have developed a radiative transport code to shoot body wave rays through general deterministic 3-D structure, including the computation of quantities required to synthesize high frequency body wave coda generated by small-scale, statistically described heterogeneity. Representing the wavefield as a sum of multiply scattered *P* and *S* waves in 3-D, it includes reflection/transmission and *P*–*S* conversion by interfaces, effects on the amplitude and polarization of scattered re-radiated waves from statistically described small-scale heterogeneity, and intrinsic viscoelastic attenuation. Velocity gradients and earth curvature can be parametrized either by thin layers and an earth-flattening transformation or by specifying velocity models at vertices of space-filling tetrahedra (see the Appendix).

Coda envelopes can be plotted in several styles: (1) as a function of time and component of motion at a single station, (2) as a function of distance and time to assist in traveltime picking and traveltime uncertainty estimates, or (3) as *Sn*apshots in time as a function of depth and range to understand the evolution of *P* and *S* energy and the homogenization of radiation pattern.

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Source radiation patterns can be input in the form of generalized moment tensors, which can include tectonic release. The effect of source spectra shape on coda envelopes can be handled by varying the scalar moment for each frequency in simulations for individual narrow band passed simulation.

The availability of both explosion and earthquake waveforms, detailed maps of regional phase efficiency, and the existence of 3-D deterministic structural models derived from global stations and transportable arrays make the Lop Nor region an ideal region to implement our modelling technique. In this region we have completed modelling experiments that have tested the effects on regional coda of parameters specifying the heterogeneity spectrum, the effects of crustal and mantle intrinsic attenuation, the effects of a thin sediment layer, the nature of the Moho transition, and the effects and detection of explosion triggered tectonic release. Tests varying the parameters specifying small-scale heterogeneity find that the strongest effects on regional codas occur when the scale length specifying the corner of a von Kármán spectrum are close to the dominant wavelength of a narrow band passed simulation. *Pn* and *Sn* amplitudes strongly depend on velocity gradients and structural complexity beneath the Moho. In narrow band simulations of coda envelopes, accurate estimates of the intensity of P and S velocity fluctuation, von Kármán scale length, and intrinsic Q in the crust are critical input to the modelling of *Lg* and the prediction of P/Lg ratios. Tectonic release detected from low frequency, teleseismic, surface waves minimally affects the high frequency (>2 Hz) regional coda in the Lop Nor region. Thus far, the most significant conclusion of all of these tests is that performance of discriminants based on P/Lg amplitude ratios is best at frequencies above 2 Hz from recordings at ranges less than 1000 km.

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APPENDIX A: CAPABILITIES

A1 Current

A1.1 Source representations

RADIATIVE3D uses a moment-tensor representation to describe seismic sources. Moment tensor representation is a truncation at quadrupole order of the multipole expansion of the elastic wave field at large distances from the source, and thus provides a way to parametrize complex sources as point-source equivalents. Despite discarding higher-order multipoles, moment tensors provide good representation of a wide range of naturally occurring and man-made seismic sources. Both idealized shear dislocation (e.g. slip on a fault plane) and idealized explosions are exactly representable by moment tensors. In fact, the six degrees of freedom in the rank-2 symmetric moment tensor allow for the specification of a full range of event types from pure isotropic (explosion/implosion), to double couple (shear-dislocation earthquake), and various other configurations such as compensated linear vector dipole, etc. Internally, the moment tensor is converted into probability distributions describing the likelihood of phonon emission at various TOAs from the event source.

A1.2 Scattering radiation patterns

A scattering event involves an interruption of the phonon ray path at a scattering location and a re-radiation of the phonon at a deflected angle. The angular dependence of scattering is determined by the scattering radiation patterns described in Sato *et al.* (2012, chap. 4). These radiation patterns depend on frequency and on four statistical parameters describing the heterogeneity, as well as on the background P and S elastic velocities. The four statistical parameters include the strength ϵ of the perturbation field, the scale length a that defines a low-pass corner in the power spectrum of length scales, a parameter κ that determines the fall-off rate of the power spectrum, and a parameter ν that establishes a ratio between velocity and density fluctuation strengths. Like event sources, the scattering shapes are also represented internally as probability distributions characterizing the relative likelihood of various deflection angles.

Our software defines a uniform scattering regime per model cell, which can be either a layer or volume enclosed by a tetrahedron. The results shown in this paper assumed models constructed from homogeneous layers. In models having linear gradients within layers or tetrahedra, both the scattering probability densities and mean

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free paths can be spatially varying. In such cases, model cells may be too large to capture the effects of the spatial variability of the heterogeneity spectrum within a cell. It may be then necessary to decrease the size of model cells to achieve a more accurate representation of scattered codas. Methods of handling spatially-varying mean free paths have been described however, e.g. Hoshiba (1994), and may guide future code enhancements to lessen this burden on the modeller.

A1.3 Flexible Earth model input

RADIATIVE3D supports Earth models based on a multitude of model construction metaphors, allowing both very simple and very complex models to be constructed with ease. An Earth-coordinate subsystem (ECS) allows model parameters, event locations, and seismometer locations to be specified in a coordinate system of the user's choice and automatic conversion to layered internal model space or tetrahedral mesh will occur. The ECS also allows Earth-flattening transformations to be applied as a user option, to simulate Earth curvature in layered models.

Full-3D Earth models are constructed from either (1) stacked cylindrical pancake cells of constant velocity, density and attenuation having planar boundaries of arbitrary tilt or from (2) space-filling tetrahedral cells in which linear model gradients are calculated from model parameters specified at the vertices of the tetrahedra.

Free-surface reflections, as well as sharp velocity discontinuities (such as at the Moho layer), require special treatment of ray propagation. Much like the radiation patterns of scattering from small-scale volumetric inhomogeneities, a ray's interaction with an interface across which seismic velocities and/or material densities abruptly change can be thought of as a form of scattering radiation pattern. For a given incident ray, there are up to four different outgoing ray directions, and two different ray types (*P* and *S*), with special care being necessary to correctly transform the *S*-polarization angle. Reflection/transmission/conversion coefficients at first-order model discontinuities (Aki & Richards 1980; Červenỳ 2001) are used by RADIATIVE3D to compute probabilities of an incoming ray transforming into the corresponding outgoing ray type. This allows for the modelling of reflection (at a free surface or interface) and transmission (interface) of rays with corresponding possibility of transformation between *P* and *S* polarization types.

A2 In development

A2.1 Anisotropy of heterogeneity scale lengths

Anisotropy of heterogeneity scale lengths that scatter high frequency seismic waves can affect coda shapes (Hong & Wu 2005) and spatial coherence of amplitudes (Tkalčić *et al.* 2010) depending on the angle their wave fronts make with the axes of vertically or horizontally stretched heterogeneity. Nielsen & Thybo (2006), for example, found the need to incorporate horizontally stretched heterogeneity to model *Pn* and *Sn* codas observed in Russian *Pn*E data. In a teleseismic problem (Cormier *et al.* 2011), we have applied some of the extensions for anisotropy of scale lengths described by Margerin (2006) and Jing *et al.* (2014) but have not yet implemented these extensions in the RADIATIVE3D code. A more efficient simulation of the effects of heterogeneity stretched in a quasi-horizontal direction might be to simply combine random perturbations to thin layering together with isotropic heterogeneity. This can be accomplished with our existing code.

A2.2 Source spectrum and earthquake slip histories

Currently RADIATIVE3D allows input of single point sources described by moment tensors with no specific options for handling the source time function. The source spectrum can be handled by applying an appropriate scalar factor to each simulated narrow band frequency envelope to simulate the effects of a source spectrum. A possible future application of radiative transport modelling will be to estimate high frequency strong ground motions excited by extended and complex earthquake slip histories. A loop over point sources distributed over a fault plane or within a volume can be added to achieve envelope simulations for spatially extended earthquake sources.

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